PLANETARY SR II — UNITED STATES

Aggregate Capacity, Stratified Substitution, and Purchased Access to Essential Function

SignalRupture (SR) | Planetary Country Study | Evidence window 2000–2025 | 2026

Scope and Evidentiary Position

This study treats the country as a coupled institutional–population system rather than a single actor or a scalar condition. The empirical window is 2000–2025 wherever comparable series permit, with later publication dates used only to recover measurements whose reference periods fall inside that window. Pressure, continuity, drift, collapse, and reorganization are evaluated by function, population, place, and duration. A deteriorating indicator is not collapse; a rising input is not capacity; and a compensatory mechanism is not presumed harmful. Official statistics, administrative records, regulatory materials, audited institutional evidence, and peer-reviewed research receive priority. Observations, SR-derived calculations, mechanism interpretations, and prospective predictions remain separately labeled. No scalar national-collapse score is constructed.

The United States is examined as a federal system with extensive state and local variation, fragmented public authority, high private-sector participation in essential infrastructures, large fiscal and technological capacity, and pronounced inequality of exposure. Its central empirical configuration is stratified substitution: households and institutions preserve function through insurance, credit, private purchase, relocation, unpaid care, employer provision, philanthropy, and local variation, while some populations delay or forgo function.

Abstract

The United States combines exceptional aggregate productive, fiscal, scientific, military, and technological capacity with highly segmented access to essential functions. The central national pattern is stratified substitution. When public or employer-linked provision is unavailable, unaffordable, geographically distant, or legally inaccessible, households and organizations purchase, borrow, relocate, self-provide, delay, or forgo. This preserves substantial national continuity while making income, employment, insurance, race, age, geography, disability, and legal status consequential to effective access.

The study applies the SignalRupture field system, Pressure Ecology theorem, cross-field bridge protocol, dependency topology, and Structural Restoration Dynamics to housing, food, household finance, healthcare, unpaid care, economic representation, demography, coercive infrastructure, communications, education, knowledge, media, algorithms, psychology, community, governance, justice, energy, logistics, climate, and disaster capacity. Official observations show both strain and countertrend. In 2023, 49.7% of renter households—more than 21 million—were housing-cost burdened. In 2024, 13.7% of households were food insecure and 5.4% experienced very low food security. In 2024, 92.0% of people had health insurance for some or all of the year, leaving about 8.0% uninsured. In 2025, 73% of adults reported doing okay financially or living comfortably and 63% could cover a $400 emergency expense with cash or its equivalent; sixteen percent did not pay all bills in the prior month and 26% skipped medical expenses because of cost. These combinations reject both generalized-collapse and aggregate-success interpretations.

The strongest bridge hypotheses connect housing cost to mobility, family formation, and health; insurance and employment to effective medical access; care demand to labour supply and household margin; digital and platform dependence to information and administrative access; coercive exposure to household, labour, and political capacity; and climate hazards to insurance, migration, municipal finance, and infrastructure. The evidence supports unequal exposure, surface–substrate divergence, substitution, and burden transfer. It does not establish a completed national collapse sequence or the predictive superiority of a unified SR model. The conditional 2031 forecast expects strain to appear first in the households, workers, caregivers, local governments, and organizations financing or performing substitute provision, followed by functional deterioration only where substitution becomes unavailable or unaffordable.

The country model also registers the transition from computational recognition infrastructure (CRI) to AI-mediated institutional substrates (AIS) and multi-agent coordination systems (MACS), together with U.S. monetary, sanctions, supply-chain, migration, platform, and technological spillovers. These are tested as directional pressure pathways rather than presumed harm. The developmental-lock-in hypothesis applies only where repeated externalization measurably weakens a receiving country's value retention, recovery capacity, substitution, or policy autonomy.

1. Replication Question and Country Boundary

The U.S. paper asks the same central question posed in Canada: does the SR grammar organize recurrent measurable relations across domains while preserving improvements, heterogeneity, and null results? Replication does not require identical indicator levels. It requires invariant construct roles. Pressure must precede or plausibly condition response; durable capacity must remain distinct from extraordinary compensation; visible output must remain distinct from effective access; collapse must be defined by a domain threshold rather than rhetoric; and reorganization must involve a durable change in who supplies function or how it is delivered.

The United States creates a hard test because national averages compress fifty states, the District of Columbia, territories, tribal jurisdictions, counties, metropolitan areas, and highly unequal markets. National continuity may coexist with local failure; local crisis may coexist with national growth. The paper therefore treats geographic and subgroup divergence as empirical content, not statistical noise.

$$P(u,d,t) \rightarrow R(u,d,t) \rightarrow D(u,d,t+h_1) \rightarrow C(u,d,t+h_2) \rightarrow O(u,d,t+h_3)$$

2. Frozen Hypotheses

Hypothesis	U.S. prediction	Failure condition
H1 Pressure–capacity divergence	Requirements or exposures outpace durable accessible capacity in multiple systems.	Comparable measures show capacity closing gaps without rising delay, burden, or substitution.
H2 Compensated continuity	Household, worker, market, community, or government compensation preserves visible function.	Compensation is unrelated to measured gaps or adds no buffering value.
H3 Surface–substrate divergence	Aggregate output or formal coverage improves while access, distribution, or resilience diverges.	Direct-condition measures move consistently with headline indicators.
H4 Institutional reflex	Responses preserve continuity, eligibility, or system legibility before closing underlying gaps.	Responses consistently eliminate generating conditions before burdens develop.
H5 Exposure Silo Bias	Restoring interacting exposures changes estimates or intervention logic.	Co-exposures produce no material inferential change in adequately powered tests.
H6 Cross-domain propagation	Earlier strain predicts later outcomes across domains after controls.	No reproducible lagged transfer appears.
H7 Visibility lag	Structural change precedes institutional recognition or response.	Recognition tracks or anticipates independently dated change.
H8 Reorganization	Persistent pressure changes the supplier, financing, geography, or mode of function.	Substitution is temporary, trivial, or unrelated to pressure.
H9 Incremental value	SR improves held-out prediction or decision utility over strong baselines.	Conventional models equal or outperform it.

3. Evidence and Method

The empirical hierarchy prioritizes federal statistics, administrative data, official regulatory evidence, and central-bank surveys. Observations from different surveys are not treated as if they describe the same respondents. Cross-domain synthesis establishes recurrence and candidate pathways; it does not manufacture individual-level linkage.

The common measurement grammar remains Q, D, $C_{\square}$, C_e, and F: effective requirement, durable capacity, sustainable flexibility, extraordinary compensation, and observable function. In the United States, effective capacity often depends upon eligibility and purchasing power. A hospital, broadband line, rental unit, insurance product, or university seat can exist while remaining unavailable to a particular population. Capacity must therefore be measured as accessible capacity rather than physical stock alone.

$$D_{eff} = D \times A \times Aff \times Elig \times Rel$$

Here A is geographic availability, Aff affordability, Elig institutional eligibility, and Rel reliability. The equation is a bookkeeping architecture, not an estimated universal multiplicative law. It prevents the existence of supply from being mistaken for effective access.

4. Housing: Capacity With Restricted Affordability

The Census Bureau reported that more than 21 million renter households spent over 30% of income on housing costs in 2023, representing 49.7% of the 42.5 million renter households for whom burden was calculated. The median renter housing-cost ratio remained 31%, but exposure varied sharply: 56.2% of Black renter households were cost-burdened, compared with different rates across other groups. The 2019–2023 five-year ACS counted 20.0 million rented units and 17.3 million owned units paying more than 30% of monthly income toward housing costs.

This is not evidence that half of renters face the same hardship. The 30% threshold is a conventional indicator, not a complete measure of residual income, local costs, household size, debt, or housing quality. It establishes a large affordability exposure and a tenure-specific divergence. Housing exists; the effective capacity to secure it without severe budget compression is unequally distributed.

The SR mechanism is stratified substitution. Households compensate through smaller units, additional earners, roommates, longer commutes, geographic relocation, delayed household formation, credit, family transfers, or reduced spending elsewhere. The national data demonstrate burden; they do not identify which compensatory pathway each household used. H1 and H3 receive descriptive support. H2 requires linked longitudinal evidence.

5. Food Security and Survival Infrastructure

USDA reported that 13.7% of U.S. households—18.3 million—were food insecure at some time during 2024. Of these, 5.4%, or 7.2 million households, experienced very low food security in which eating patterns were disrupted and intake was reduced. The 2024 prevalence was not statistically different from 13.5% in 2023 or 12.8% in 2022, but it was significantly higher than rates observed from 2016 through 2021. Rural households had a 15.4% food-insecurity rate in 2023 compared with 13.5% nationally.

The finding is persistence after deterioration, not continuous acceleration. Food insecurity is a direct-condition measure with a defined resource basis. It coexists with large agricultural capacity, extensive retail systems, federal nutrition programs, charitable provision, and positive GDP growth. That coexistence strongly supports Surface–Substrate Divergence: national productive capacity is not identical to secure household access.

A decisive Exposure Silo Bias test would model housing burden, income volatility, disability, rurality, household composition, benefit access, healthcare costs, and transportation jointly. The present aggregates show interacting exposures are plausible but do not prove that restoring them changes any particular coefficient.

6. Household Financial Resilience

The Federal Reserve's Survey of Household Economics and Decisionmaking reported that 73% of adults in 2024 said they were doing okay or living comfortably financially, below the 78% high in 2021 but broadly similar to recent years. Sixty-three percent said they would cover a hypothetical $400 emergency expense using cash, savings, or a credit card paid off at the next statement. That share was also unchanged from recent years and below 68% in 2021.

The result is neither generalized destitution nor full resilience. Roughly three in eight adults would not cover the emergency expense entirely with cash or its equivalent under the survey definition. The same national system therefore contains a majority with immediate $400 capacity and a large minority relying on borrowing, selling, partial payment, or inability to cover the expense.

Metropolitan and nonmetropolitan differences reinforce the distributional interpretation. In 2024, 64% of metropolitan adults and 57% of nonmetropolitan adults reported cash-equivalent capacity for the $400 expense. Parents living with children under 18 reported 55%, compared with 65% among other adults. These gaps support exposure stratification without establishing a closed debt loop for every household.

Financial indicator	Observed condition	SR judgment
Doing okay / comfortable	73% of adults in 2024	Majority continuity; below 2021 high
$400 cash-equivalent capacity	63% in 2024	Large resilient majority and substantial exposed minority
Metro / nonmetro	64% versus 57%	Geographic exposure difference
Parents / other adults	55% versus 65%	Household-structure exposure difference
Causal debt loop	Not identified by these aggregates	Not established

7. Healthcare: Coverage Improvement and Effective-Access Segmentation

The United States differs structurally from Canada because insurance, employment, public programs, provider networks, deductibles, prices, geography, and eligibility jointly shape access. CDC reported that the all-age uninsured rate fell from 9.7% in 2020 to 8.2% in 2024. The 2024 estimate represented 27.2 million uninsured people. Almost two-thirds of people under age 65 had private coverage and more than one-quarter had public coverage.

This is an important restoration result. Coverage improved; Planetary SR must score it as improvement. At the same time, universal effective access was not achieved. Coverage is not identical to affordability, network availability, appointment access, or absence of medical debt. Historical CDC evidence also shows that cost-related delayed or forgone care varies sharply by insurance, disability, race, and region, demonstrating that formal coverage and effective access are distinct constructs.

Healthcare therefore supports H3 in a bounded form: coverage improvement coexists with residual uninsurance and access stratification. It does not establish healthcare collapse. Nor does it yet provide a matched national Q–D–C–F panel comparable to the Canadian hospital staffing analysis. The U.S. healthcare test is strongest as a segmentation and restoration case, not a compensatory-inversion case.

8. Unpaid Care and the Household Capacity Layer

BLS estimated that 14% of the civilian noninstitutional population aged 15 or older—38.2 million people—provided unpaid eldercare during the combined 2023–2024 period. On an average day, 28% of eldercare providers supplied care, spending 3.9 hours on caregiving days. Non-employed providers averaged 4.8 hours and employed providers 2.8 hours on days they provided care.

This is direct evidence that households and communities supply a large capacity layer. It is not direct evidence that government transferred the entire burden. Care may reflect preference, relationship, reciprocity, culture, and family responsibility. The stronger SR claim requires measurement of formal-care availability, cost, eligibility, unmet need, intensity, caregiver consequences, and replacement capacity.

The current result supports the existence and scale of compensatory or complementary care. It partially supports H2 and H8, because a durable supplier outside formal institutions is observable. It does not establish exhaustion or a collapse transition.

9. Economic Representation

BEA reported real GDP growth of 2.8% in 2024. Retail trade, health care and social assistance, and professional, scientific, and technical services were among leading contributors. This demonstrates strong aggregate productive continuity. It does not resolve housing affordability, food security, emergency savings, healthcare access, or correctional exposure because GDP measures production rather than distribution or effective access.

The U.S. evidence therefore reproduces the metric-boundary result found in Canada without reproducing the same real-GDP-per-capita pattern. Aggregate growth coexisted with material insecurity, but the paper does not assert that most direct-condition measures deteriorated in the same period. The correct conclusion is that GDP cannot by itself adjudicate population condition. Surface–Substrate Divergence is observed as coexistence, not as a claim that the surface indicator is wrong.

The negative control matters: 73% of adults reported doing okay or living comfortably, health insurance coverage improved, and several justice indicators remained below decade-earlier levels. A theory that converted positive GDP into hidden universal decline would fail. The evidence is mixed by design.

10. Demographic Reproduction and Recovery

CDC recorded 3,628,934 births in 2024. The number rose 1% from 2023, while the general fertility rate declined 1% from 54.5 to 53.8 births per 1,000 women aged 15–44. The apparent difference is not contradictory: population composition and the number of women at risk can allow births to rise while the rate falls. It demonstrates why demographic counts and rates must remain distinct.

The United States is not classified as demographically collapsed. Low fertility may affect future cohort replacement, but migration, productivity, labour-force participation, health, care infrastructure, regional mobility, and family adaptation modify consequences. The national test identifies reproduction pressure, not demonstrated loss of recovery capacity.

The U.S. case differs from Canada because the present paper does not quantify migration dependence as the dominant growth source. It therefore preserves the common construct while withholding a country-specific inference not established by the selected evidence.

11. Coercive Infrastructure and Quiet Governance

Bureau of Justice Statistics data show that 5.63 million adults were under correctional supervision at year-end 2023. Approximately 3.77 million were on probation or parole and 1.85 million were incarcerated in prison or jail. The prison population was 1,254,200, 2% above 2022. Yet males serving more than one year were 21% below 2013 and females were 18% below 2013. Local jails held 664,200 people at midyear 2023, and the jail incarceration rate of 198 per 100,000 residents was 14% lower than a decade earlier.

This is a mixed trajectory: short-run increases coexist with longer-run declines. The scale of community supervision is structurally important because coercive governance extends beyond confinement into reporting, eligibility, surveillance, compliance, fees, and reentry conditions. However, counts do not by themselves establish collapse, injustice, or discriminatory mechanism.

Collapse Criminology's strongest U.S. application would test whether formal legal status and visible non-incarceration coexist with sustained administrative and economic burden among supervised populations, and whether exposure propagates into employment, housing, health, and family outcomes. The current national evidence establishes the infrastructure's scale and reorganization toward community supervision; causal propagation remains pending.

12. Communications and Digital Infrastructure

The FCC's Broadband Data Collection and Communications Marketplace reporting make availability, provider presence, performance, subscription, pricing, and geographic gaps measurable. The Thirteenth Measuring Broadband America report found a weighted average advertised download speed of 467 Mbps among participating providers, 52% above the prior report. This is capacity improvement, not deterioration.

National performance improvement does not automatically produce universal effective access. Subscription depends on price, local competition, installation, device access, digital skills, reliability, and household circumstances. Tribal, rural, low-income, and remote communities may face different substitution conditions. The current paper does not assign a national dependency or concentration score because the available official summary does not identify one validated cross-market coefficient.

The infrastructure result is therefore mixed: technical capacity improved, while the structure remains suitable for testing stratified access and government dependence. GAUID's stronger agency-loss claim is not established.

13. Education, Media, Knowledge, and AI

Education is both a service domain and a recovery mechanism. The United States has extraordinary university, research, and training capacity, but elementary and secondary performance is decentralized across states and districts. The 2022 NAEP cycle documented broad post-pandemic losses, and federal analysis found that 22% of fourth-grade public-school students reported five or more absences in the preceding month—twice the 2019 share. The proper SR interpretation is not that schools ceased functioning. It is that attendance, learning recovery, staffing, household time, disability support, and local finance jointly determine whether formal enrolment becomes effective learning.

The education buffer is distributed among teachers, aides, parents, tutors, community organizations, and private markets. Families with money and flexible time can purchase tutoring, relocate, change schools, or supervise recovery. Families without those options absorb learning disruption directly. This is a strong candidate for stratified substitution, but the national aggregates do not identify the sequence required for a collapse claim. The 2031 registry therefore monitors chronic absenteeism, teacher vacancies, parent time, achievement recovery, and subgroup gaps separately.

Media, knowledge, and AI are treated as distinct systems rather than collapsed into a single epistemic score. Media conditions require measures of local-news supply, platform dependence, information diversity, trust, and correction. AI requires deployment-specific evidence about inherited data, decision authority, error, appeal, and subgroup effects. Federal agencies more than doubled reported AI use from 2023 to 2024, according to GAO, while reporting and governance remained uneven. Scale is evidence of institutional transition; it is not evidence that automated decisions are uniformly harmful or that human agency has already disappeared.

13.1 Research capacity and expert integration

The United States remains a global research and innovation center. This durable capacity is an important negative control against total-decline narratives. Yet research production and institutional use are different constructs. Publication, patenting, and private investment do not guarantee that local governments, schools, hospitals, courts, utilities, or households can translate knowledge into timely function. SR therefore measures expert integration at the decision boundary: whether evidence changes resource allocation, whether implementation is staffed, whether feedback is incorporated, and whether failure can be acknowledged without institutional penalty.

The main risk is not an absence of expertise but selective translation. High-capacity organizations can purchase analytics, legal advice, cybersecurity, and specialized labor. Smaller jurisdictions and low-margin institutions may face vacancies, procurement delay, weak data, or vendor dependence. A national average for research spending would conceal this conversion gap. The prediction is conditional: where translation capacity falls while requirements rise, human improvisation and contracted substitution should increase before visible output deteriorates.

13.2 Media and public legibility

Public legibility is the capacity to determine what is happening, who is responsible, and what remedy is available. The U.S. information environment combines strong investigative institutions and open-data capacity with fragmented audiences, platform mediation, local-news contraction, strategic communication, and rapidly changing synthetic media. These conditions can increase search and verification costs without implying that the population has entered a uniform epistemic collapse.

The SR test is behavioral. If institutional signals become less trusted or less usable, people should rely more heavily on peer networks, paid intermediaries, partisan identity, or withdrawal. That substitution may keep decisions moving while increasing unequal error exposure. The stronger claim would be falsified if trust and information-use measures improve, local verification capacity expands, and corrections propagate faster even as platform use grows.

13.3 AI and administrative decision systems

The governing principle is system inheritance. Automated systems inherit categories, objectives, omissions, and appeal structures from the institutions that commission them. They may improve consistency and throughput; they may also scale an existing eligibility rule or create a new burden of proof. The relevant question is not whether an agency uses AI, but whether the system changes effective access, contestability, workload, and error distribution.

GAO’s 2025 review found agencies already managing generative-AI use cases while reporting challenges involving data, workforce, policy, and acquisition. OMB’s 2025 guidance preserved annual use-case inventories and public reporting. This is an active reorganization of administrative capacity. The 2031 forecast does not presume net harm: it predicts that automation without independent appeal and staffing will shift effort toward claimants and frontline workers, whereas automation paired with audit, explanation, and human review can restore capacity.

14. Psychology, Wellbeing, and Effective Agency

Psychological condition is not treated as an isolated pathology. Effective agency depends upon time, money, predictable rules, safety, health, social support, and credible routes to remedy. A person may remain employed and housed while losing discretionary time, tolerating chronic uncertainty, delaying care, or narrowing future plans. Those changes are potential early indicators of buffer use because they can occur before institutional output visibly fails.

The United States also supplies a restoration signal. Life expectancy at birth rose from 78.4 years in 2023 to 79.0 in 2024. That improvement must be scored as recovery, not explained away. It does not eliminate large subgroup differences or prove broad psychological recovery, but it shows that a valid SR model must accommodate reversal. Population wellbeing should therefore be monitored through multiple outcomes—distress, suicide, overdose, disability, sleep, loneliness, and perceived control—without converting any one measure into a national verdict.

The forecastable mechanism is constraint accumulation. Rising obligations and weaker predictability can reduce perceived control; reduced control can increase withdrawal, conflict, short-term decision horizons, and reliance on immediate substitutes. The causal chain remains a hypothesis until panel evidence separates selection, macroeconomic shocks, health conditions, and institutional effects. The strongest prediction is temporal: agency indicators should deteriorate among highly exposed groups before later exits, service loss, or institutional performance decline.

15. Family, Community, and Social Cohesion

Families and communities are an operating layer of the U.S. system. The 38.2 million unpaid eldercare providers measured in 2023–2024 are one visible component. Households also coordinate transport, childcare, disability support, school recovery, navigation of insurance and benefits, disaster response, and reentry after incarceration. These activities create real function. They should not be romanticized as costless resilience.

Community capacity is unevenly renewable. Dense social networks, wealth, flexible employment, stable housing, and local organizations can distribute a shock. Repeated shocks can instead concentrate responsibility in the same caregivers and volunteers. The distinction between solidarity and exhaustion is empirical: whether participation is voluntary, whether substitutes exist, whether health and employment consequences rise, and whether the load can rotate without loss of function.

Cohesion is likewise differentiated. Conflict at the national level may coexist with high local trust; local solidarity may coexist with exclusion; declining institutional trust may coexist with confidence in particular professions or organizations. The national paper therefore rejects a single trust score. It predicts that unresolved burden plus weak remedy will intensify boundary formation—between insured and uninsured, owners and renters, secure and precarious workers, protected and exposed places—unless institutions visibly restore reciprocity.

16. Governance, Administrative Burden, and Political Capacity

American governance is polycentric. Federalism, separation of powers, courts, tribal sovereignty, state and local authority, private contractors, and nonprofit delivery create redundancy and experimentation, but also veto points and navigation costs. The same architecture can prevent centralized failure and make responsibility difficult to locate. SR treats fragmentation as neither inherently resilient nor inherently dysfunctional; its effect depends on coordination, fiscal capacity, staffing, and remedy.

Administrative burden is one mechanism by which institutional pressure becomes human work. Learning costs, compliance costs, documentation, waiting, appeals, repeated identity proof, and uncertainty can shift capacity requirements to applicants, families, professionals, and local offices. Digitization can reduce this burden or merely move it to an interface. The decisive measure is completed access with error correction, not portal availability.

The institutional-buffer prediction is specific. Agencies facing rising caseloads and limited durable capacity will first rely on overtime, vacancy tolerance, contractor labor, queue management, narrower eligibility interpretation, self-service, and delayed maintenance. Visible outputs can remain stable while error, turnover, appeals, and user effort rise. Once those compensatory channels weaken, delays and discontinuities should become more visible. If agencies expand durable staffing and simplify rules while burdens fall, the forecast should be redirected toward restoration.

17. Criminal Justice, Rights, and Effective Remedy

Formal rights are not equivalent to usable remedy. Effective remedy requires information, counsel, time, money, language access, physical access, procedural reliability, and enforcement. The United States has extensive constitutional, statutory, judicial, and civil-society capacity, yet its use is stratified by jurisdiction and resources. This is a central national-system domain because unresolved disputes can propagate into housing, employment, family stability, health, and political legitimacy.

Correctional data show a mixed rather than monotonic trajectory. At year-end 2023, 5.63 million adults were under correctional supervision, including 1.85 million incarcerated and 3.77 million supervised in the community. Prison and jail levels remained below decade-earlier measures even as the prison population rose from 2022. Community supervision represents a durable reorganization: confinement is partly replaced by reporting, monitoring, conditions, fees, service requirements, and reentry administration.

The SR question is whether this reorganization restores agency or transfers hidden burden. A favorable pathway would show declining revocation, improved employment and housing, accessible counsel, and reduced family disruption. An adverse pathway would show formal decarceration combined with high compliance load, technical violations, debt, and cross-domain exclusion. Both are testable. Counts alone cannot decide between them.

18. Energy, Transportation, Logistics, and Critical Infrastructure

The United States is resource-rich and technologically capable, but its infrastructures are tightly interdependent. Electricity supports communications, water, healthcare, finance, fuel, logistics, and public administration. Transportation supports labor access and household survival as well as commerce. National abundance can coexist with local fragility where redundancy, affordability, maintenance, and emergency restoration are weak.

Electricity generation reached a record 4.43 thousand terawatt-hours in 2025, 2.8% above 2024. This is a strong capacity signal. At the same time, demand growth from manufacturing, electrification, and data centers changes planning requirements after a long period of relatively flat demand. EIA projects material server load growth, and regional systems face different generation, transmission, water, and permitting constraints. The correct diagnosis is expansion under a changing load profile, not grid collapse.

Transportation and logistics reveal the access filter. A job, clinic, school, or food outlet can exist yet remain ineffective without reliable travel. Households compensate through longer commutes, multiple vehicles, delivery services, informal rides, relocation, or forgone trips. Commercial systems compensate through inventories, rerouting, supplier diversification, and price. These adaptations can be resilient, but repeated cost transfer to drivers, warehouse workers, ports, utilities, and households is a candidate human-buffer pathway.

Cyber dependence adds a cross-domain multiplier. Recovery depends not only on technical defense but on staffing, procurement, vendor concentration, exercises, offline procedures, and truthful incident reporting. A cyber event becomes an SR propagation event only when loss of one service creates measurable downstream loss in others. The 2031 registry therefore separates incident frequency from restoration time and human workload.

19. Climate, Ecology, and Disaster Capacity

Climate pressure is already recurrent and spatially uneven. NOAA identified 27 U.S. weather and climate disasters exceeding $1 billion in losses during 2024, with total costs of approximately $182.7 billion. These measures exclude many smaller events and do not represent a full welfare estimate, but they establish repeated large shocks across a system that must restore housing, infrastructure, insurance, health, education, and local government simultaneously.

The United States possesses extraordinary emergency, engineering, fiscal, philanthropic, and private-insurance capacity. Yet recovery is filtered by tenure, wealth, insurance availability, documentation, credit, local tax base, disability, and migration options. An affluent household may insure, rebuild, relocate, or litigate; an exposed renter or small municipality may experience prolonged displacement. Climate therefore amplifies the national mechanism of stratified substitution.

The human buffer appears in responders, utility crews, clinicians, teachers, claims workers, relatives, volunteers, and displaced households. The key forecast is not that a single disaster collapses the country. It is that repeated shocks shorten recovery intervals. Where restoration remains incomplete before the next shock, overtime, household savings, insurance capacity, and local staffing should deteriorate first; infrastructure reliability and institutional continuity should worsen later. Major investment in mitigation, insurance reform, housing resilience, and local capacity is a direct falsifying or restorative pathway.

20. Cross-Domain Synthesis

20.1 The recurring U.S. relation

The U.S. pattern can be expressed as high gross capacity combined with stratified effective access. Housing stock, food production, healthcare expenditure, digital bandwidth, financial markets, and national output are extensive. Yet access depends upon income, tenure, insurance, geography, eligibility, household structure, and legal position. When universal provision is incomplete, people compensate through purchase, credit, unpaid work, relocation, community provision, delayed consumption, or forgoing the function.

Gross Capacity → Eligibility / Affordability / Geography Filters → Unequal Effective Access → Substitution or Forgoing

20.2 Comparison with Canada

Dimension	Canada calibration	United States replication
Dominant pattern	Compensated continuity under public-system pressure	Stratified substitution under high aggregate capacity
Healthcare	Universal formal coverage with access and staffing gaps	Coverage segmentation with improvement but residual uninsurance
Household buffer	Essential-cost compression and revolving credit	Emergency-expense resilience split by geography and family status
Care	Large unpaid child/adult care layer	Large unpaid eldercare layer
Demography	Low fertility, aging, migration-supported growth	Low fertility pressure; births rose while fertility rate fell
Coercive infrastructure	Not integrated in first paper	Large incarceration and community-supervision system; mixed trend
National collapse	Not supported	Not supported

20.3 Why the replication matters

The United States does not merely repeat Canadian findings. The same grammar identifies a different institutional conversion mechanism. Canada more often preserves universal formal structures through compensation and delay. The United States more often filters access through segmented public and private arrangements, generating substitution by purchasing power, eligibility, employment, geography, and household resources. This is evidence of framework flexibility without claim flexibility: the constructs remain fixed while their national configuration changes.

20.4 Field-System Architecture

The country model is a field-preserving system. Each domain i has a state vector $x_i(t)$ containing variables appropriate to that domain—such as access, delay, affordability, staffing, reliability, effective remedy, household margin, or ecological exposure. Within-field dynamics are represented as $x_i(t+1) = F_i x_i(t) + G_i z_i(t) + \varepsilon_i(t)$, where z_i contains observed pressures and interventions. The equation is a model class, not an estimated national law. It prevents a finding in one field from silently becoming evidence in another.

A cross-field claim is registered only through an explicit bridge: $x_i(t) \rightarrow M_{i\square}(t, \tau_{i\square}) \rightarrow x_{\square}(t+\tau_{i\square})$. $M_{i\square}$ must identify the transmitting exposure or institution, temporal order, transformation, expected lag $\tau_{i\square}$, amplifiers, buffers, reverse direction, common shocks, and a falsifier. The stacked country system is $X(t+1) = AX(t) + GZ(t) + L\Omega(t) + \varepsilon(t)$. A represents preregistered domestic bridges; Ω represents common, transnational, or planetary shocks. Estimating A is a future empirical task. The present papers register bridge hypotheses and grade the evidence supporting each link.

Pressure Ecology is used as a theorem rather than counted as a twenty-fourth field. It asks how pressure is generated, imported, absorbed, transformed, transferred, externalized, displaced in time, and propagated. For domain k, a bookkeeping identity is $P^*_{\square}(t+1) = P^*_{\square}(t) + G_{\square}(t) + I_{\square}(t) - A_{\square}(t) - E_{\square}(t) - D_{\square}(t) + \Sigma_{\square}T_{\square\rightarrow\square}(t) - \Sigma_{\square}T_{\square\rightarrow\square}(t)$. P* is residual effective pressure; G is domestic generation; I is imported pressure; A is genuine absorption; E is externalization beyond the boundary; D is temporal displacement; and T is cross-domain transfer. Transformation changes the form of pressure and is recorded in the relevant bridge rather than treated as disappearance. Displaced pressure must be re-entered when its obligation matures.

The field inventory is therefore neither a checklist nor a claim that all twenty-three fields are equally activated. A field is diagnostic in a country paper only when the country evidence identifies its distinct object, measures at least one state variable, specifies a mechanism, defines a prediction, and states a falsifier. “Contextual,” “indeterminate,” and “not diagnostic” are valid outcomes. The three specialized subfields—the SR Matrix Field, the SR Policy Substrate Field, and Structural Restoration Dynamics—organize integration, intervention, and recovery without erasing the autonomy of the full fields.

The canonical field count used in this study is twenty-three full fields plus three specialized subfields. Pressure Ecology, the Cross-Field Integration Framework, DVS, equations, indices, operators, and diagnostic systems are canonical components but are not counted again as fields.

20.5 Pressure Ecology Ledger

The U.S. pressure ecology is distinguished by market-mediated transformation. Pressure is frequently converted into prices, premiums, deductibles, debt, travel, eligibility work, private contracting, unpaid care, or geographic sorting. These mechanisms can be adaptive and innovative, but their availability is stratified. The country can therefore sustain strong output and institutional continuity while a minority experiences direct functional loss and a larger group carries the cost of substitution.

Process	Country mechanism	Evidentiary judgment
Generation	Housing costs, healthcare prices, care demand, inequality, violence, climate exposure, administrative fragmentation	Multiple direct measures; mechanisms vary by state and group.
Importation	Trade and supply chains, migration, energy markets, cyber/platform shocks, global finance	High global coupling with substantial domestic buffering capacity.
Absorption	Federal fiscal capacity, insurance, employers, private capital, state programs, household wealth	Large but unequally available.
Transformation	Need becomes price, premium, debt, travel, waiting, eligibility work, or forgone use	Central national mechanism; observable in several domains.
Transfer	Public → private; employer → worker; state → locality; institution → household/community	Strong descriptive evidence; causal quantities require linked data.
Externalization	Environmental, labour, incarceration, and supply-chain burdens shifted across places/groups	Plausible and partly measured; boundary choice decisive.
Temporal displacement	Consumer/student/medical debt, deferred maintenance, delayed care, postponed formation	Observable; long-run consequences heterogeneous.

Propagation	Housing–health–demography; care–labour; climate–insurance–migration; justice–employment	Priority bridges with mixed causal maturity.
Restoration	Universal or targeted access gains, lower burden, reduced coercive exposure, resilient infrastructure	Several partial gains; strong state heterogeneity.

The ledger distinguishes disappearance from relocation. Absorption reduces effective pressure; transformation changes its form; transfer changes its bearer; externalization changes the analytic boundary; temporal displacement changes when it becomes visible. A country can therefore look stable while residual pressure accumulates elsewhere in the coupled system. Conversely, a high level of compensation can represent successful and sustainable institutional flexibility. Classification depends on duration, voluntariness, replacement cost, recovery, and whether the generating gap closes.

20.6 Field Activation Matrix

The matrix records where each canonical field has a distinct country-level object. “Primary” indicates direct relevance and at least one measurable state; “Bridge” indicates that the field is most useful at an interface; “Contextual” indicates that present national evidence does not justify an independent mechanism claim. These labels govern this paper only; they are not judgments about the field’s status elsewhere.

No.	Field or subfield	Role	Country object
1	Institutional Cognition	Primary	Visibility, representation, federal/state/local recognition.
2	SignalRupture Institutional Governance Field (SR-IGF)	Primary	Fragmented authority, response, eligibility, burden allocation.
3	Collapse Governance	Bridge	Private, household, and local compensation for institutional gaps.
4	The Drift Collapse Field (DCF)	Primary	Domain thresholds amid high aggregate continuity.
5	Systems Propagation Science (SPS)	Primary	Housing–health–care–demography and climate–insurance pathways.
6	SR Cyber-Epistemic Infrastructural Diagnostic Field (SR-CEIDF)	Bridge	Cloud, telecom, cyber, identity, platform dependency.
7	The SR Algorithm Field	Bridge	Eligibility, employment, credit, policing, ranking, and appeals.
8	Infrastructural Political Economy	Primary	Ownership, labour, housing, finance, market-mediated access.
9	Infrastructural Overshoot	Primary	Climate, energy, logistics, settlement, insurance and renewal.
10	System Inheritance Studies (SIS)	Bridge	Legacy rules inherited by digital and privatized systems.
11	Collapse Epistemology	Bridge	Media fragmentation, trust, ranking and public legibility.
12	SignalRupture Research Incentive Field (SR-RIF)	Bridge	Federal, university, philanthropic and commercial research incentives.
13	Collapse Psychology	Primary	Constraint, debt, insecurity, care and exposure.
14	Collapse Sociology	Primary	Stratification, community infrastructure, segregation and institutional burden.

15	SR Anthropology	Primary	Race, migration, legal status, family and local adaptation.
16	Collapse Demography	Primary	Fertility, mortality, migration, household formation, regional recovery.
17	Childhood Cognitive Infrastructure Studies (CCIS)	Primary	Material co-exposures, schools, childcare and digital access.
18	Infrastructural Medicine	Primary	Insurance, affordability, provider capacity, effective access.
19	Collapse Criminology	Primary	Incarceration, supervision, violence, community alternatives.
20	Post-Web Jurisprudence	Primary	Effective remedy, automated evidence, privacy and platform power.
21	Collapse Religion	Contextual	Contextual: congregation and mutual aid; no single national collapse mechanism.
22	Collapse Economics	Primary	Margin, debt, housing, market substitution, output–condition divergence.
23	Collapse Society	Primary	Country coupling without totalization.
24	The SR Matrix Field	Bridge	Cross-domain matrix and heterogeneous state/local interfaces.
25	The SR Policy Substrate Field	Primary	Instrument–substrate alignment across federal and state variation.
26	Structural Restoration Dynamics	Primary	Recovery, burden reduction, reversals and negative cases.

Activation is claim-specific. A “Primary” label does not validate every proposition associated with a field, and a “Contextual” label does not disconfirm the field. The empirical burden remains at the level of construct, mechanism, bridge, population, and prediction.

20.7 Cross-Field Bridge Register

The following bridges are the country model’s priority causal candidates. Each is directional for testing even where feedback is expected. Status describes the present evidence, not inevitability.

Source field	Mechanism/exposure	Receiver	Lag	Status and falsifier
Housing/Collapse Economics	Cost burden reduces margin and constrains mobility	Health; demography; labour	1–7 years	High descriptive; test policy and metro variation
Healthcare/Infrastructural Medicine	Employment, insurance, network, and price mediate effective access	Household finance; health; labour	Immediate–5 years	Strong segmentation; coverage is not utilization
Care/SR Anthropology	Unpaid and paid care substitute for fragmented provision	Labour; household margin; psychology	Immediate–10 years	Supported prevalence; causal burden varies
Food/Collapse Economics	Price and income shocks change food access and coping	Childhood; health; psychology	Immediate–5 years	Annual measurement supports timing tests
Justice/Collapse Criminology	Supervision and incarceration alter employment, family, and civic access	Households; demography; governance	Months–20 years	Mechanism literature strong; national SR increment open

Algorithms/SIS	Private and public systems inherit eligibility, credit, employment, and policing rules	Jurisprudence; governance; economics	Immediate–3 years	Requires decision audit; no universal AI amplification claim
Media/Collapse Epistemology	Platform ranking and fragmented media alter common visibility	Institutional cognition; governance	Days–4 years	Competing media explanations required
Climate/Overshoot	Hazards raise insurance, infrastructure, health, and migration loads	Housing; municipal finance; labour	Immediate–15 years	Strong physical basis; recovery pathway local
Research/SR-RIF	Funding, publication, and commercial incentives direct knowledge production	Policy; algorithms; medicine	2–10 years	Test against independent agenda and uptake measures
Digital/SR-CEIDF	Cloud, telecom, identity, and platform concentration create common dependencies	Commerce; government; epistemics	Immediate–5 years	Dependency clear; failure probability separately tested

No bridge is supported merely because its source and receiver indicators both deteriorate. The minimum design requires temporal order, a transmitting exposure, an independently measured receiver, common-shock controls, reverse-path testing, and a negative or recovery case. Where a bridge cannot meet this contract, it remains a registered hypothesis rather than part of the empirical conclusion.

20.8 Dependency and Substitution Topology

National capacity depends not only on the quantity of resources but on whether critical nodes can be substituted under stress. For node m, the heuristic dependency relation is $R_{\square} = \kappa_{\square}\chi_{\square}\delta_{\square}/(1 + \sigma_{\square})$, where κ is functional criticality, χ is provider or pathway concentration, δ is jurisdictional or geographic distance from effective control, and σ is tested substitution capacity. $R_{\square}$ is not estimated here and is not a country score. It states what must be measured before concentration is interpreted as fragility.

Critical function	Dependencies	Substitution capacity	Pressure implication
Health access	Employer insurance, public programs, networks, providers, pricing	High for some; low for uninsured or thin markets	Substitution is stratified by income, work, state, and health status
Housing	Credit, zoning, construction, insurance, local infrastructure	Low in constrained metros over short horizons	Relocation transfers costs to commuting, networks, and opportunity
Food	Income, logistics, retail, assistance programs	Moderate; uneven in rural and low-access areas	Programs and charity absorb some pressure without eliminating income constraints
Digital platforms	Cloud providers, broadband carriers, app ecosystems, identity/payment rails	Technically possible but costly at scale	Common-mode dependence coexists with high innovation capacity
Electric grid	Regional operators, generation, transmission, fuel, weather	Regionally variable	Interconnection adds resilience and correlated exposure
Care	Household time, paid services, employers, state programs	Low for many low-income households	Unpaid care is a critical hidden dependency
Local government	Property tax, state rules, federal transfers, bond markets	Unequal by tax base and legal authority	Fiscal substitution varies sharply by jurisdiction
Disaster recovery	Insurance, FEMA, state/local capacity, contractors, household wealth	High for wealthy insured groups; lower elsewhere	Recovery averages can conceal displacement and non-return

Substitution is evaluated by speed, coverage, affordability, quality, legal authority, and reversibility. A substitute that exists only for high-income households or major metropolitan institutions is not national substitution. A local or informal substitute is counted as real capacity, but its labour, health, legal, ecological, and opportunity costs remain part of the system ledger.

20.9 Structural Restoration Dynamics and Negative Cases

Restoration is a change in the generating substrate, not simply a rebound in the surface indicator. The operational condition is $\Delta G < 0$ together with stable or improving function and declining dependence on extraordinary compensation. Compensatory inversion is the contrary warning pattern: $G > 0$, $\Delta C_e \leq 0$, and subsequent $\Delta F < 0$, where G is the requirement–capacity gap, C_e extraordinary compensation, and F observed function. The present evidence does not establish a national compensatory inversion.

Domain	Countertrend or restoration evidence	Interpretation
Health insurance	92.0% had coverage for some or all of 2024	Coverage improvement is a real countertrend; cost-related non-use remains a separate outcome.
Household finance	73% doing okay/living comfortably and 63% able to cover $400 in 2025	Rejects generalized financial collapse; subgroup declines and unpaid bills remain.
Justice	Prison and jail rates remain below earlier peaks	Rejects monotonic coercive expansion; distribution and community supervision still matter.
Employment/output	Strong aggregate output and large innovative capacity persist	A genuine capacity measure, but not a proxy for universal access.
Food security	Most households remained food secure in 2024	Negative case against national survival-system failure; 13.7% insecurity remains consequential.
Education/research	Large universities, research systems, and expert networks persist	Rejects epistemic-system collapse; access, incentives, and translation remain testable.
Digital access	Broadband and device access expanded	Access gains do not eliminate quality, affordability, platform, or rural gaps.
Disaster recovery	Federal and market capacities can restore assets and services	Restoration must include return, affordability, and reduced recurrent residual load.

Negative cases are part of the test rather than rhetorical balance. A valid national model must explain why some domains improve, why some exposed places avoid deterioration, why some policies reduce burden, and why the same pressure does not produce the same result everywhere. These cases identify buffers, threshold differences, institutional learning, and omitted variables. If the model cannot accommodate them without changing definitions after observation, the integrated claim fails.

20.10 DVS Validation and Robustness Tests

The Diagnostic Validation Standard (DVS) separates the existence of a field, the evidence for a country claim, and institutional recognition. The country model proceeds through operationalization, measurement validation, longitudinal or comparative testing, strong conventional controls, mechanism testing, adversarial falsification, replication, and intervention or restoration testing. The following checks govern interpretation of this manuscript.

Check	Application	Status
Source validation	Census 2024 coverage: 92.0% insured for some or all of the year; USDA 2024 food insecurity: 13.7%.	Confirmed
Current household check	Federal Reserve 2025 SHED: 73% doing okay/comfortable; 63% able to cover $400; 16% unpaid bills; 26% skipped medical expenses due to cost.	Confirmed
State heterogeneity	National averages cannot identify Medicaid, housing, justice, energy, or climate mechanisms without state/local models.	Required
Surface/substrate separation	GDP and innovation capacity retained as real capacity but not treated as universal access.	Applied
Negative cases	Coverage gains, high food security, lower justice rates than earlier peaks, and stable household indicators retained.	Applied

Conventional baselines	Welfare-state, labour, health economics, federalism, segregation, market power, and climate-risk models estimated first.	Required
Sensitivity	Results must survive alternate burden thresholds, subgroup definitions, state effects, lag windows, and survey sources.	Prospective
Forecast lock	2031 test requires buffer or substitute strain to precede a defined functional loss in the same exposed unit.	Falsifiable

Incremental value is the decisive comparison. For each outcome y, the strongest mature disciplinary model B is estimated first; SR variables and registered bridges are added only afterward. The claim survives only if B+SR improves held-out prediction, calibration, warning lead time, mechanism discrimination, or intervention choice. In notation: $\Delta V = V(B+SR) - V(B)$. A non-positive, unstable, or non-replicating ΔV weakens the relevant SR claim even when the descriptive synthesis remains useful.

Robustness requires alternative denominators, thresholds, lags, regional units, population weights, missing-data assumptions, and bridge directions. Results must be reported with confidence or uncertainty intervals where estimable. Post-hoc index weights are prohibited. Measurement changes and publication lags are recorded as possible visibility effects but cannot be treated as proof of institutional concealment.

20.11 Integrated Country Judgment

The U.S. pressure ecology is distinguished by market-mediated transformation. Pressure is frequently converted into prices, premiums, deductibles, debt, travel, eligibility work, private contracting, unpaid care, or geographic sorting. These mechanisms can be adaptive and innovative, but their availability is stratified. The country can therefore sustain strong output and institutional continuity while a minority experiences direct functional loss and a larger group carries the cost of substitution.

The field system adds value at the level of disciplined integration: it makes burden bearers, substitutions, dependencies, temporal displacement, and recovery conditions explicit across domains. It does not replace the mature literatures that identify domain mechanisms, and it does not convert simultaneous pressures into one causal story. The national finding is therefore a bounded topology of continuity and exposure. Its strongest causal and predictive claims remain registered for panel, quasi-experimental, and held-out testing.

20.12 AI Substrate Transition: CRI, AIS, and MACS

The AI environment is modeled as a substrate transition rather than a standalone technology sector. Computational recognition infrastructure (CRI) detects and classifies patterns. AI-mediated institutional substrates (AIS) place those systems inside eligibility, triage, pricing, enforcement, forecasting, allocation, procurement, scheduling, and remedy. Multi-agent coordination systems (MACS) arise when interacting automated agents operate across institutions, firms, households, platforms, and countries. Each phase can increase capacity while also inheriting the country's existing categories, dependencies, inequalities, and continuity mechanisms.

The United States is likely to scale AI through employers, insurers, platforms, finance, health systems, government contractors, and state and local administrations at unequal speeds. High-capacity households and institutions may obtain capable substitution first, while access for others remains tied to employment, insurance, subscription, data, geography, or platform status. The core national test is whether AI reduces stratification or simply makes stratified substitution faster and less visible.

Phase	Country insertion	Registered prediction	Decisive test
CRI	Pattern recognition in credit, insurance, employment, health, policing, logistics, and public administration	Faster classification and risk sorting	Prediction gains persist after subgroup calibration and selection controls
AIS	AI embedded in allocation, pricing, eligibility, surveillance, appeals, and service delivery	Purchased access and institutional capacity diverge further	Access and remedy improve for low-capacity groups, not only premium users
MACS	Interacting corporate, governmental, financial, logistics, and household agents	High coordination with correlated platform, data, and market failures	Provider exit or model failure does not disable multiple essential functions

2031 test	AI capability diffuses faster than rights, contestability, and universal access	Surface productivity rises before household margin and effective access	Distributional access, workload, appeals, and dependency improve together

The five-year prediction is directional and conditional. AI should initially extend human and organizational buffers by accelerating documentation, triage, forecasting, and coordination. Surface throughput may therefore improve before effective access, household margin, workload, rights, or outcomes. As delegation deepens, cloud and model providers become critical dependency nodes, institutional visibility may lag behind operational change, and human-buffer exhaustion may reappear as exception handling, monitoring, appeals, and responsibility without authority. The AI claim is supported only if these sequences outperform conventional technology-adoption, labour, organizational, and inequality models in held-out tests.

AI counts as structural restoration only when it reduces the generating requirement–capacity gap while maintaining or improving function and lowering extraordinary workload, exclusion, dependency, and transferred burden. If AI improves headline throughput while those substrate measures worsen, it is classified as compensation, temporal displacement, or pressure transfer—not restoration.

20.12.1 Locked Forecast Ladder, 2026–2031

The 2031 endpoint is retained as the final adjudication date, but it is not the first evaluation date. The following checkpoints are registered prospectively so an acceleration during 2028–2029 can be detected without moving the forecast after observation. Each checkpoint tests a stage in the predicted sequence rather than requiring a national crisis or uniform outcome.

Checkpoint	**Model stage**	**Country observation**	**Decision rule**
2026–2027	Baseline formation	Record AI access by income, employment, insurance, geography, platform status, and institutional capacity together with workload, appeals, pricing, eligibility, and provider concentration.	Baseline is frozen before later distributional outcomes are interpreted.
2028	Early divergence	Productivity and service throughput should improve first among high-capacity institutions and purchasers, before universal gains in effective access, remedy, or household margin.	Broad gains among low-capacity groups weaken the stratified-substitution prediction.
2029	Acceleration test	Delegation in credit, insurance, employment, health, logistics, and administration should make automated exception, denial, appeal, oversight, and platform-dependency burdens more visible.	Comparable adoption without widening burden or dependency weakens the predicted sequence.
2030	Propagation test	AI-mediated differences should propagate across employment, insurance, credit, healthcare, education, public benefits, and local institutional capacity rather than remain sector-specific.	No reproducible cross-domain order after state, market, and common-shock controls rejects propagation.
2031	Adjudication	Classify the observed trajectory as broad restoration, faster stratified substitution, or intensified platform dependency.	Restoration requires distributional access, workload, remedy, household margin, and substitution capacity to improve together.

Temporal compression is recorded when the registered sequence appears earlier than expected. Earlier timing does not validate later stages automatically; each mechanism, bridge, distributional effect, and restoration test must still be observed independently. Conversely, failure of the 2028 or 2029 leading indicators under substantial adoption weakens the forecast before 2031. The model therefore cannot preserve itself by postponing evaluation to the final year.

20.13 International Pressure Externalization and Developmental Lock-In

The international ledger records a directional vector $\Omega^{d}_{(c\to r)}(t)$, where c is the sending country, r the receiving country, and d the domain. It is multidimensional and cannot be collapsed into a single externalization score. Every registered pathway must identify the origin, mechanism, receiver, benefit bearer, cost bearer, lag, substitution or refusal capacity, return pathway, evidence, and falsifier. Ordinary interdependence is not equivalent to asymmetric pressure transfer, and asymmetric transfer is not automatically developmental destabilization.

The United States is a central generator and transmitter of monetary, financial, sanctions, technology, platform, supply-chain, security, and migration pressure. Mexico, Guatemala, Honduras, and El Salvador are priority North American and Central American receivers through labour demand, supply chains, migration enforcement, agriculture, and remittance dependence. Cuba, Venezuela, and Iran are distinct sanctions and transaction-cost cases; vulnerable emerging markets are exposed to dollar funding and U.S. monetary shocks. These pathways can also produce benefits, and domestic policy or mismanagement in receiving states remains a competing explanation.

Receiver or corridor	Pressure pathway	Benefit and cost incidence	Evidence and falsification test
Mexico; Guatemala; Honduras; El Salvador	Supply chains, labour demand, migration enforcement, agriculture, security cooperation, and remittance dependence	U.S. firms and consumers gain labour and supply capacity; migrants, households, and origin institutions bear variable costs	Test wages, value retention, fiscal capacity, return conditions, coercive exposure, remittance use, and local substitution
Cuba; Venezuela; Iran	Sanctions, financial exclusion, transaction costs, asset and trade restrictions	Policy leverage and domestic political goals versus household, firm, and public-system burden abroad	Separate sanctions effects from domestic governance, commodity shocks, conflict, and third-country intervention
Vulnerable emerging markets	Dollar funding, interest-rate transmission, capital flows, risk repricing, and demand shocks	U.S. monetary stabilization can shift qualification costs to foreign borrowers and governments	Estimate heterogeneous responses by balance-sheet exposure, reserves, institutions, and policy response
Global AI and digital supply chain	Model control, cloud concentration, minerals, electricity, water, data labour, moderation, and administrative standards	U.S. firms and users capture rents and capability while physical and labour burdens distribute internationally	Trace ownership, contracts, resource load, labour conditions, downstream liability, and provider substitution

Developmental destabilization is registered only where the pathway persistently weakens a receiving country's fiscal, institutional, ecological, labour-reproductive, technological, or recovery capacity. The developmental-lock relation is $L_r(t+1) = f[X_r, H_r, D_r, E_r, C_r] - g[V_r, S_r, R_r, B_r]$. X is externally captured value; H human-capacity drain; D debt or foreign-exchange exposure; E ecological residual load; C institutional capture or constraint; V domestic value retention; S substitution; R restoration; and B bargaining and regulatory capacity. This is a test architecture, not an estimated index.

The hypothesized lock sequence is: external extraction or policy pressure → reduced domestic value retention → ecological, labour, fiscal, or institutional burden → weakened recovery and substitution → increased dependence on foreign capital, markets, technology, or migration → reduced bargaining capacity → renewed externalization. Confirming evidence would include high extraction or labour export with weak value retention, loss of trained capacity, unrepaired ecological liabilities, foreign-currency or platform dependence, stalled structural transformation, weak fiscal or regulatory capacity, and repeated recovery failure. The hypothesis weakens when similarly exposed countries improve productivity, diversification, public revenue, local infrastructure, labour conditions, ecological restoration, substitution, and policy autonomy.

AI adds a new externalization channel. Model ownership, cloud control, rents, and standards may concentrate in one jurisdiction while electricity, water, minerals, data work, content moderation, and downstream administrative risk are distributed elsewhere. By 2031, the registered prediction is that AI will increase both cross-border coordination and common-mode dependency. It will contribute to developmental lock-in only where receiving countries lose value retention, substitution, remedy, or bargaining capacity; diffusion that builds sovereign or locally governed capability is a direct negative case.

20.14 Canonical Planetary Country Protocol

Every Planetary SR country paper uses the same fixed analytic sequence so national differences are comparable without forcing them into a single rank or collapse score.

Stage	Required module	Minimum country-paper content

1–3	Boundary, baseline, and surface/substrate separation	Define country system, period, populations, mature disciplinary baselines, and the difference between output and lived or institutional capacity.
4–6	Field activation, pressure ecology, and human buffer	Activate only diagnostic fields; trace generation, import, absorption, transformation, transfer, externalization, displacement, propagation, and compensation.
7–9	Bridges, dependencies, and unequal exposure	Register directional cross-field mechanisms, critical nodes, substitution, geography, institutions, households, firms, communities, and populations.
10–11	Restoration and negative cases	Distinguish substrate repair from rebound or compensation; retain improvements, resistant cases, and failed predictions.
12	AI substrate transition	Assess CRI, AIS, and MACS; inheritance, workload, access, contestability, providers, common-mode failure, and 2031 predictions.
13	International pressure ledger	Name sender, receiver, mechanism, benefit bearer, cost bearer, lag, refusal, return path, evidence, and falsifier.
14	Developmental lock-in test	Test value retention, human-capacity drain, debt/FX exposure, ecological residuals, institutional constraint, restoration, substitution, and bargaining.
15	Validation and forecast lock	Estimate conventional baselines first; test incremental value, sensitivity, causal order, uncertainty, preregistered predictions, and falsification.

The protocol prohibits deterministic country labels. No country is classified as doomed, permanently developing, collapsed, or causally responsible for another country's condition from exposure data alone. Conclusions remain pathway-specific, time-bounded, distribution-sensitive, and revisable only through stated evidence and falsifiers.

21. National Scorecard

Proposition	U.S. evidence	Judgment
Pressure–capacity divergence	Housing, food, health and household resilience show access gaps despite large capacity.	Supported descriptively
Compensated continuity	Unpaid eldercare and financial substitution are observable; full gap linkage incomplete.	Partially supported
Surface–substrate divergence	GDP, broadband and insurance improvements coexist with persistent unequal access.	Supported
Institutional reflex	Programs and segmented markets preserve function, but sequencing is not uniformly identified.	Indeterminate / partial
Exposure Silo Bias	Multiple interacting exposures specified; unified re-estimation not performed.	Not yet decisively tested
Cross-domain propagation	Housing–food–health–finance and supervision pathways plausible; causal lags unestimated.	Not achieved
Visibility lag	No common onset-recognition dataset assembled.	Not achieved

Proposition	U.S. evidence	Judgment
Reorganization	Private purchase, insurance segmentation, unpaid care, relocation and community supervision are durable architectures.	Supported as structure; causal transition partial
Incremental predictive value	No national held-out comparison against strong baselines.	Not achieved
United States as collapse state	Large capacity, restoration cases, mixed trends and no unified threshold.	Not supported

22. Alternative Explanations and Falsifiers

Several mature explanations compete with the SR synthesis. Poverty and wealth models may explain access differences without an additional framework. Federalism may explain geographic variation. Labor-market segmentation, racial stratification, administrative burden, social determinants of health, market power, and disaster vulnerability each provide domain-specific mechanisms. SR earns value only if its frozen constructs improve integration, temporal prediction, or intervention choice beyond these baselines. It is not validated by redescribing their findings.

- Linked household data show that housing burden, food insecurity, medical access, emergency savings, and substitution are largely independent after strong controls.
- Effective access differences disappear when availability, affordability, eligibility, geography, and reliability are measured consistently.
- Unpaid eldercare intensity is unrelated to formal-care gaps or caregiver outcomes.
- Stratified access variables add no explanatory or predictive value beyond established poverty, insurance, geography, and administrative-burden models.
- Cross-domain transfers fail to replicate in state, county, household, or institutional panels.
- Restoration of coverage, affordability, or capacity fails to reduce compensatory burden where the architecture predicts it should.
- Strong conventional models consistently outperform the frozen SR augmentation on held-out data.

23. Limitations

- National aggregates combine deeply heterogeneous state, local, tribal, territorial, and market systems.
- Different surveys do not observe the same people or institutions.
- The paper contains descriptive replication rather than a harmonized longitudinal causal model.
- Housing-cost burden is an incomplete measure of residual income and housing adequacy.
- Food insecurity, financial resilience, insurance, unpaid care, and correctional exposure use different reference populations.
- Broadband performance among participating providers cannot establish nationwide affordability or competition.
- Correctional counts do not identify fairness, deterrence, harm, or discriminatory mechanism.
- Education, media, epistemic and deployed-AI modules remain incomplete.
- No validated scalar Planetary SR score or national collapse threshold is used.

24. Decisive U.S. Test

The next stage is a state-year and county-year panel joined to household and institutional microdata where lawful and feasible. The United States offers greater cross-sectional variation than Canada, but that advantage creates a risk of specification searching. Outcomes, exposure blocks, lags, and holdout states must therefore be frozen before model selection.

$$Y_{(s,d,t+h)} = \alpha + \rho Y_{(s,d,t)} + \beta P_{(s,d,t)} + \gamma Access_{(s,d,t)} + \delta Comp_{(s,d,t)} + \theta X_{(s,d,t)} + \mu_{\square} + \lambda d + \tau t + \varepsilon_{(s,d,t+h)}$$

1. Estimate strong domain baselines before adding SR terms.

2. Model availability, affordability, eligibility, geography, and reliability separately before any composite.
3. Use state policy variation and credible natural experiments where identification assumptions hold.
4. Test whether compensation buffers function, predicts later burden, or falls during unresolved pressure.
5. Restore cross-domain exposures in preregistered blocks to test Exposure Silo Bias.
6. Hold out states, counties, periods, or institutions for genuine predictive comparison.
7. Score restoration, nulls, and reversals with the same rules used for deterioration.

25. Five-Year Forecast: Human Buffer Precedence

Forecast horizon: August 2026 through August 2031. The forecast is conditional, not prophetic. It asks whether unresolved capacity gaps first consume the people and organizations supplying extraordinary compensation, and whether visible institutional deterioration follows. The United States should not be judged by a single national event. Its most likely pathway is rolling and stratified: exhaustion appears first in exposed populations and low-margin institutions while high-capacity groups continue to purchase substitutes.

The U.S. forecast freezes the same governing sequence used in Canada:

Institutional capacity gap → human compensation → visible continuity → cumulative human burden → human-buffer exhaustion → compensatory inversion → visible institutional deterioration

The national mechanism differs in its conversion step. Canadian continuity is more often buffered beneath universal public structures. U.S. continuity is more often maintained through credit, additional paid work, unpaid care, private purchase, insurance, relocation, philanthropy, contractors, and unequal local capacity. Consequently, national averages may remain stable while buffer exhaustion becomes severe in particular states, counties, occupations, racialized groups, renters, disabled people, caregivers, and disaster-exposed communities.

$$G_{\square} = Q_{\square} - (D_{\square} + C_{\square,\square})$$

$$G_{\square} > 0 \text{ and } \Delta H_{\square} > 0 \Rightarrow F_{\square} \text{ may remain provisionally stable}$$

$$G_{\square} > 0 \text{ and } \Delta H_{\square} \leq 0 \Rightarrow P(\Delta F_{\square+\square} < 0) \text{ increases}$$

Here G is the unresolved gap between effective requirement Q and durable capacity plus sustainable flexibility; H is available human compensatory capacity; and F is observable function. These are structural relations. Their empirical proxies are frozen by domain below and must not be substituted after outcomes are known.

25.1 Domain predictions to 2031

Domain	Human-buffer indicators expected first	Later institutional indicator	2031 directional prediction
Households / housing	Emergency savings, debt service, extra earners, doubling-up, involuntary moves	Eviction, homelessness, arrears, shelter and voucher pressure	Unequal depletion; renters and low-income families lead
Healthcare / care	Overtime, vacancy duration, unpaid-care hours, delayed care, medical debt	Waits, closures, missed care, continuity loss	Local and occupational deterioration before national failure
Education	Teacher absence/turnover, parent time, tutoring purchase, chronic absenteeism	Learning recovery, course access, special-education delay	Private substitution widens recovery differences
Government / justice	Caseworker load, vacancies, contractor use, appeals, claimant effort	Backlogs, error, remedy delay, supervision failure	Administrative burden rises before visible discontinuity
Infrastructure / cyber	Crew overtime, deferred maintenance, vendor dependence, manual workarounds	Outage and restoration time, service interruption	Restoration intervals lengthen in low-capacity places

Domain	Human-buffer indicators expected first	Later institutional indicator	2031 directional prediction
Climate / disaster	Responder fatigue, insurance withdrawal, household savings loss, displacement	Incomplete recovery, municipal fiscal stress, repeated service loss	Compounding shocks expose stratified recovery capacity
Information / AI	Verification work, appeal effort, frontline review, expert vacancies	Uncorrected error, exclusion, trust and uptake loss	Outcome depends on audit and human-review capacity

25.2 Evaluation rule

The national forecast is supported only if at least two pre-specified human-buffer indicators deteriorate before or during later institutional deterioration in at least three monitored domains. Timing must be established with annual or finer data, and the relationship must survive domain baselines, state and period effects, composition checks, and plausible policy confounding. A national headline alone cannot satisfy the rule.

The forecast is strongly supported if buffer decline improves out-of-sample prediction of later function beyond conventional measures of income, unemployment, population, policy, and prior outcomes. It is partially supported if the temporal order replicates but adds little predictive value. It is not supported if institutional deterioration generally precedes buffer decline, if buffers replenish while function worsens, or if durable capacity expansion reduces both burden and failure without the predicted inversion.

25.3 Competing 2031 pathways

Pathway	Observable pattern	Interpretation
Restoration	Durable staffing, affordability, simplification, resilience investment and appeal capacity rise; human burden falls; function improves.	SR restoration; prediction of exhaustion does not occur
Buffered continuity	Function remains stable, compensation remains high but renewable, and health/turnover do not deteriorate.	Pressure without exhaustion
Stratified inversion	Exposed groups lose savings, time, health or staffing first; local failure follows while aggregate capacity remains high.	Primary U.S. prediction
Direct institutional shock	Policy, fiscal, cyber, conflict or disaster shock damages function before measurable human-buffer decline.	Human-buffer precedence falsified for that episode
National synchronized failure	Multiple domains cross functional thresholds together across most jurisdictions.	Low-probability, high-consequence pathway; not the base case

25.4 Frozen forecast judgment

Base case: by 2031 the United States is more likely to display stratified compensatory inversion than nationwide synchronized collapse. Human buffers are likely to be used up in particular populations, occupations, and jurisdictions, not uniformly across the country. High-income households, large firms, wealthy jurisdictions, and high-margin institutions will continue purchasing substitutes longer. The visible institutional phase will therefore appear as closures, backlogs, insurance withdrawal, staff turnover, absenteeism, delayed remedy, longer restoration, and geographic exit—distributed unevenly and often normalized as separate local problems.

This is the crucial qualification. 'The human buffer collapses first' is retained as a testable precedence claim, but it is no longer stated as total national exhaustion. The U.S. prediction is rolling exhaustion under unequal exit options. If measurement shows broad replenishment of time, savings, health, staffing, and trust while institutional function deteriorates independently, the claim fails.

26. Final Empirical Judgment

Planetary SR survives its first replication. The United States does not reproduce Canada mechanically, and that is the important result. The frozen architecture identifies a distinct national configuration: high aggregate productive and infrastructural capacity filtered through affordability, eligibility, insurance, geography, tenure, household structure, and legal position. The result is stratified effective access and widespread substitution outside universal public provision.

The evidence strongly supports Surface–Substrate Divergence and unequal exposure. It partially supports compensated continuity through unpaid care, financial substitution, segmented insurance, relocation, and other household or market responses. It shows real restoration: health insurance coverage improved, broadband performance increased, and several correctional indicators remained below decade-earlier levels. These findings prevent SR from converting every observation into decline.

The paper does not establish that the United States is a collapse state. It does not prove a completed national pressure-to-collapse sequence, cross-domain causal propagation, or predictive superiority. Its strongest valid diagnosis is: high national capacity with stratified effective access, persistent material insecurity for substantial populations, and reorganization through private, household, geographic, and supervisory systems.

27. What the United States Establishes

The United States establishes that Planetary SR can preserve national variation without surrendering its common protocol. The country is not a larger Canada. Its dominant structure is high aggregate capacity filtered through affordability, eligibility, employment, insurance, geography, tenure, and legal position. That architecture creates stratified effective access and a larger role for market purchase, credit, household work, relocation, contractors, and charitable provision.

It also establishes that human-buffer analysis must be distributional. A national buffer can appear intact because burden is concentrated and substitutes remain available to powerful actors. The correct unit is therefore person–institution–place over time. The forecast becomes more precise: exposed human systems should show declining renewal capacity before later local institutional deterioration, while protected systems may continue to function or improve.

28. Conclusion

Canada showed how visible continuity can be supported by compensatory capacity beneath formal public systems. The United States shows something adjacent but not identical: visible abundance and output can coexist with access filters that require households to purchase, borrow, self-provide, relocate, qualify, wait, or forgo. Planetary SR can therefore distinguish two national pathways without changing its core grammar.

The U.S. replication also disciplines the Multiverse. Positive GDP growth is real. Health coverage improved. Broadband performance increased. Birth counts rose even while the fertility rate fell. Prison and jail trends were mixed rather than monotonically worsening. A valid planetary theory must preserve those countercurrents.

The result is neither national-collapse rhetoric nor conventional indicator reporting. It is a structural diagnosis of how capacity becomes—or fails to become—effective access. The United States strengthens Planetary SR at the operational and comparative levels. The decisive frontier remains causal and predictive: whether frozen SR variables improve out-of-sample explanation, warning, and intervention beyond the mature literatures from which its components are drawn.

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Appendix A — U.S. Construct Registry

Domain	Requirement / pressure	Capacity / compensation	Outcome	Status
Housing	Affordable adequate shelter	Stock, income, subsidy, sharing, relocation	Cost burden and stability	Pressure supported
Food	Reliable resource access	Income, programs, charity, household qualification	Food security	Persistent exposure
Finance	Routine and emergency obligations	Income, savings, credit, asset sale, family support	Emergency resilience	Stratified capacity

Domain	Requirement / pressure	Capacity / compensation	Outcome	Status
Healthcare	Coverage and timely affordable care	Private/public insurance, providers, household payment	Insurance and effective access	Restoration plus segmentation
Care	Eldercare requirement	Formal services and unpaid household care	Care completion and caregiver load	External capacity observed
Demography	Cohort reproduction and adaptation	Migration, productivity, participation, care	Recovery capacity	Pressure; collapse unproven
Coercive	Justice-system caseload and supervision	Prison, jail, probation, parole, legal access	Custody, supervision, reintegration	Scale and reorganization observed
Digital	Reliable affordable connectivity	Networks, providers, public programs, household devices	Subscription and performance	Capacity improvement; access test pending
Education / epistemic / AI	Learning, information and decision integrity	Institutions, households, platforms, audit and appeal	Learning, trust, error and inherited disparity	Not yet integrated

Appendix B — Cross-National Replication Rule

Canada remains the country case and the United States is the first replication, not a recalibration. The grammar, hypotheses, status categories, and falsifiers remain frozen. Country-specific measures may differ, but their construct role must be documented. No country is classified as collapsed without domain thresholds, duration, population, and functional failure. Improvements and restoration are mandatory evidence, not exceptions to hide. The next country should maximize institutional contrast and data quality rather than similarity.

Appendix C — 2031 Human Buffer Forecast Registry

Registry date: 24 August 2026. Horizon close: 24 August 2031. The entries below are frozen before future outcomes are observed. Indicator definitions may be changed only for documented discontinuities in an official series; the construct, direction, and ordering rule may not be changed.

ID	Domain	Pressure proxy	Human-buffer proxy	Function proxy	Expected order
US-HB-01	Housing	Rent-to-income and residual-income burden	Emergency savings, doubling-up, moves, debt	Eviction, homelessness, shelter demand	Buffer worsens before function
US-HB-02	Health	Demand, acuity, insurance and provider gap	Overtime, vacancies, unpaid care, delayed care	Missed care, closure, waiting, continuity	Buffer worsens before function
US-HB-03	Education	Enrolment needs, attendance and recovery gap	Teacher turnover, parent time, tutoring	Achievement, course and support access	Buffer worsens before function
US-HB-04	Administration	Caseload, rule complexity, claims and appeals	Vacancies, contractor use, applicant hours	Backlog, error and benefit discontinuity	Buffer worsens before function

ID	Domain	Pressure proxy	Human-buffer proxy	Function proxy	Expected order
US-HB-05	Justice	Caseload and supervised population	Counsel load, family support, compliance time	Delay, revocation, remedy and reentry	Buffer worsens before function
US-HB-06	Infrastructure	Load, asset age, incidents and severe weather	Overtime, deferred maintenance, workarounds	Outage and restoration duration	Buffer worsens before function
US-HB-07	Disaster	Repeated hazard and insured-loss exposure	Responder load, savings, insurance availability	Displacement and incomplete restoration	Buffer worsens before function
US-HB-08	AI / information	Automated decisions and information volume	Review, verification and appeal effort	Error correction, uptake and trust	Buffer worsens before function

Minimum support: three domains satisfy the registered order, each with at least two deteriorating human-buffer measures, and the pattern survives preregistered controls. Strong support additionally requires held-out predictive gain. Failure: fewer than three domains satisfy the order, institutional decline systematically comes first, or durable-capacity restoration improves outcomes without the predicted buffer depletion.

Appendix D — Interpretation Guardrails

- Do not infer national collapse from a large harmed population when aggregate and regional functions remain heterogeneous.
- Do not treat private purchase or household work as inherently illegitimate; measure whether it is voluntary, affordable, renewable, and effective.
- Do not combine distinct domains into a scalar national collapse score.
- Do not substitute new indicators after results are known merely to preserve the prediction.
- Do not interpret correlation between burden and failure as proof of temporal precedence.
- Score improvements, durable-capacity additions, null results, and reversals with the same weight as deterioration.
- Use identical measures and periods before making numerical Canada–United States comparisons.